

LESSON 9-1 • SESSION 2 • PREREQUISITE PRACTICE A • SKILL 1 OF 2

Multiply by One-Digit Numbers

Builds on Grade 4 • 4.NBT.B.5

TODAY'S LESSONYou will fill a table for 18 cartons in each crate, like 18×5 for 5 crates.**YOU NEED TO**Multiply a two-digit number by a one-digit number, like 18×5 . **You will use it in Practice Form A, problems 3 and 4.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Split** the two-digit number into tens and ones.
- 2 Multiply** each part by the one-digit number.
- 3 Add** the two products.

WATCH ME*the steps in action* **35×3** **Step 1** $35 = 30 + 5$ **Step 2** $30 \times 3 = 90$. $5 \times 3 = 15$.**Step 3** $90 + 15 = 105$ **Answer 105****TRY IT WITH HELP***fill in each step***1.** 18×5 **Step 1** $18 = 10 + \underline{\hspace{2cm}}$ **Step 2** $10 \times 5 = \underline{\hspace{2cm}}$. $8 \times 5 = \underline{\hspace{2cm}}$.**Step 3** $50 + 40 = \underline{\hspace{2cm}}$ **2.** 25×6 **Step 1** $25 = \underline{\hspace{2cm}} + 5$ **Step 2** $20 \times 6 = \underline{\hspace{2cm}}$. $5 \times 6 = \underline{\hspace{2cm}}$.**Step 3** $120 + 30 = \underline{\hspace{2cm}}$ **YOUR TURN***use the steps on your own***3.** Multiply. Use the same three steps.

a. $24 \times 3 = \underline{\hspace{2cm}}$

b. $45 \times 4 = \underline{\hspace{2cm}}$

c. $28 \times 5 = \underline{\hspace{2cm}}$

d. $16 \times 7 = \underline{\hspace{2cm}}$

e. $36 \times 4 = \underline{\hspace{2cm}}$

f. $30 \times 8 = \underline{\hspace{2cm}}$

Find the Rule in a Table

Builds on Grade 4 • 4.OA.C.5

TODAY'S LESSONYou will find the rule for x : 1, 2, 3 and y : 12, 24, 36, then find y when $x = 7$.**YOU NEED TO**Find the rule that turns each x into y , like "multiply by 12." **You will use it in Practice Form A, problem 7.****STEPS TO FOLLOW***use these same steps on every problem*

- 1** Look at the **first** pair. Try add and try multiply.
- 2** **Test** each rule on the other pairs.
- 3** **Write** the rule that works for every pair.

WATCH ME*the steps in action* **x : 1, 2, 3 y : 7, 14, 21****Step 1** 1 to 7: add 6, or multiply by 7.**Step 2** Add 6: $2 + 6 = 8$, not 14. Multiply by 7: $2 \times 7 = 14$, $3 \times 7 = 21$.**Step 3** The rule is multiply by 7.**Answer** **multiply by 7****TRY IT WITH HELP***fill in each step***4.** x : 1, 2, 3 y : 15, 30, 45**Step 1** 1 to 15: add _____, or multiply by _____.**Step 2** Multiply by 15: $2 \times 15 =$ _____, $3 \times 15 =$ _____.**Step 3** The rule is _____.**5.** x : 1, 2, 3 y : 7, 8, 9**Step 1** 1 to 7: add _____, or multiply by _____.**Step 2** Add 6: $2 + 6 =$ _____, $3 + 6 =$ _____.**Step 3** The rule is _____.**YOUR TURN***use the steps on your own***6.** Write the rule: "multiply by ____" or "add ____."**a.** x : 1, 2, 3 y : 11, 22, 33 _____**b.** x : 1, 2, 3 y : 25, 50, 75 _____**c.** x : 3, 4, 5 y : 13, 14, 15 _____**d.** x : 1, 2, 3 y : 9, 18, 27 _____**e.** x : 2, 3, 4 y : 6, 7, 8 _____**f.** x : 1, 2, 3 y : 30, 60, 90 _____**READY CHECK***a problem like the one in today's lesson***7.** A table shows x : 1, 2, 3 and y : 8, 16, 24. What is y when $x = 6$?**A** 14**B** 32**C** 48**D** 30**CHECK YOURSELF**
☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____